2. The following table is for $(1 + \log x)$. Determine estimates of $d(1 + \log x)/dx$ at $x = 0.15, 0.19, 0.23$ using (a) one term, (b) two terms, and (c) three terms of Eq. (1.4). By comparing to the analytical values, determine the errors of each estimate.

x	$1 + \log x$	Δ	Δ^2	Δ^3
0.15	0.1761			
		0.0543		
0.17	0.2304		−0.0059	
		0.0484		0.0009
0.19	0.2788		−0.0050	
		0.0434		0.0011
0.21	0.3222		−0.0039	
		0.0395		0.0006
0.23	0.3617		−0.0033	
		0.0362		0.0005
0.25	0.3979		−0.0027	
		0.0335		0.0002
0.27	0.4314		−0.0025	
		0.0310		0.0005
0.29	0.4624		−0.0020	
		0.0290		
0.31	0.4914			

3. Write expressions for the errors in each computation of Problem 2, by properly interpreting Eq. (1.6). From these expressions find upper and lower bounds for each of the computations.

4. If one wished the derivative of $(1 + \log x)$ at $x = 0.29$, the table in Problem 2 would be inadequate for this, using Eq. (1.4), because the necessary forward differences cannot be computed. A formula in terms of backward differences can be derived, however, by differentiating the Newton-Gregory backward interpolation polynomial. Do this, and use three terms to obtain $d(1 + \log x)/dx$ at $x = 0.29$ from the data given in table of Problem 2.

5. Derive the error term for the formula of Problem 4, and show that the "next-term rule" applies.

Section 2

6. Use the symbolic method to derive a formula for (a) the third derivative and (b) the fourth derivative of $f(x)$ at $x = x_0$ from a difference table.

7. The Newton-Gregory backward interpolating formula can be developed by expanding the symbolic relation

$$f_s = E^s f_0 = (1 - \nabla)^{-s} f_0.$$

Using this representation of the interpolation formula, derive the derivative formula required in Problem 4 by the symbolic technique.